

SignalWeave

Project description

Hossam Elshahaby

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ABB challenge: FROM WEAK SIGNALS TO PROACTIVE DECISIONS

Purpose

SignalWeave is an industrial investigation prototype that brings measurements, maintenance notes, alarms and video evidence into one workspace. It helps an operator inspect weak signals, understand the evidence and decide whether an action deserves review.

Problem

A small rise in motor current may appear harmless in isolation. A temperature trend or repeated operator report can change its meaning. When information sits in separate systems, teams spend time assembling context and may overlook a developing issue. Video can add useful evidence, but detecting objects alone does not establish whether an incident is imminent.

Solution and intended users

The intended users are maintenance engineers, operations teams and warehouse safety reviewers. The proposed workflow links a suspected issue to its supporting evidence, offers an investigation or action for human review, and records the eventual outcome. The prototype remains independent of ABB and has no demonstrated production integration.

App capabilities

Data import: CSV measurements and operational context provide an entry point for investigation.

Video evidence: paired RGB MP4 and object_detection.jsonl files support annotation checks and a ground-truth overlay.

Benchmarks: object detection uses separate predictions and reference boxes. A reviewed event editor supplies event intervals for early-warning evaluation.

Action workflow: investigation, action review and outcome tracking form the intended operational loop.

Expected impact

The intended impact is less time gathering evidence, clearer prioritization of inspections and more traceable decisions. For maintenance, the application could connect measurement trends with operator observations. For warehouse safety, it could help reviewers inspect worker and forklift interactions. These are proposed benefits. No measured reduction in downtime, incidents or cost has been established.

Evaluation and current evidence

The development demo uses NVIDIA PhysicalAI synthetic warehouse scenes. Matching object annotations supply classes and 2D boxes per frame. They do not supply reviewed near-miss start and end times. Run metadata also does not replace event labels.

Object-detection runs have appeared in the app. The latest supplied early-warning screenshot shows one saved event label, zero recorded chronological analyses and zero matched analyzed clips. Early-warning performance therefore remains unscored in that evidence. Clip classification also requires independently reviewed class labels and completed analyses. Synthetic detection results cannot establish performance in a real warehouse.

Benchmark rules

Detection: report precision, recall and F1 alongside the matching rule, sampled frames and exclusions.

Early warning: compare timestamped warnings with reviewed event intervals for the exact same clip. Report matched events, misses, false warnings and lead time under a declared matching policy.

Classification: freeze the positive-class mapping and compare clip predictions with independently reviewed classes.

Keep evaluation labels separate from model inputs. Preserve frame positions when annotations are incomplete. Never convert predictions into ground truth.

Demo sequence

1. Open the app and load a video with its matching object annotations.
2. Check the run and camera identity, frame count and annotation problems.
3. Reveal the reference boxes and demonstrate detection scoring.
4. Review the video and save a genuine event interval, if the event is visible.
5. Complete chronological analysis for that exact clip, then score warning timing.
6. Show prerequisites honestly if analysis or labels are still missing.

Next steps

Complete and verify the chronological-analysis flow. Add direct creation of reviewed clip classes. Persist video and analysis records consistently. Expand testing to multiple runs and normal scenes. Validate with real industrial data, then assess operational outcomes with a pilot partner.

Links

Live app: <https://snap-and-see-pro.lovable.app>

Dataset: NVIDIA PhysicalAI WorldModel Synthetic Warehouse Operations Scenes